

Assignment 2 Solution

- 1) Opcode specifies the type of instruction and what operation the Cpu will perform
- 2) The funct3 field will define the size of the data

3) a.

```
ld x5, 32(x20)
ld x6, 56(x21)
beq x5, x6, Outer_else
bne x5, x0, Inner_else
addi x29, x5, 2
sd x29, 32(x20)
addi x22, x22, -5
beq x0, x0, Exit
```

```
Inner_else:
srli x28, x6, 4
sd x28, 56(x21)
beq x0, x0, Exit
```

```
Outer_else:
sll x7, x6, 2
add x7, x7, x6
sd x7, 56(x21)
addi x22, x22, 10
```

```
Exit:
```

b.R-Type (add x7,x7,x21)

xxxxxxx	10101	00111	xxx	00111	xxxxxxx
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I-Type (addi x22,x22,-5)

11111111011	10110	xxx	10110	xxxxxxx
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S-Type (sd x29,32(x20))

0000001	11101	10100	xxx	00000	xxxxxxx
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- 4) 0000000 01011 01010 000 01001 0110011
Add x9, x10, x11

0000000 00101 10001 101 10010 0010011
Srli x18, x17, 5